

Jing-Zhang Urban AI Testbed: The First Mile from Lab to City

City-in-the-Loop: affected users and operators determine revision, scale, or exit

Rail is not one line: three systems, three operating conditions

Underground HSR and visible Line 13 coexist in the middle; HSR returns to grade in the north

FORMER SURFACE JING-ZHANG / NON-OPERATING HERITAGE

Surface remnants, removed sections, and public-space opportunities

JING-ZHANG HSR / OPERATING

Beijing North at grade → Qinghuayuan Tunnel → returns to grade in the north

METRO LINE 13 / OPERATING

Viaduct, embankment, and surface; visible does not mean crossable

Strategic diagnosis: concentrated innovation, uneven everyday support

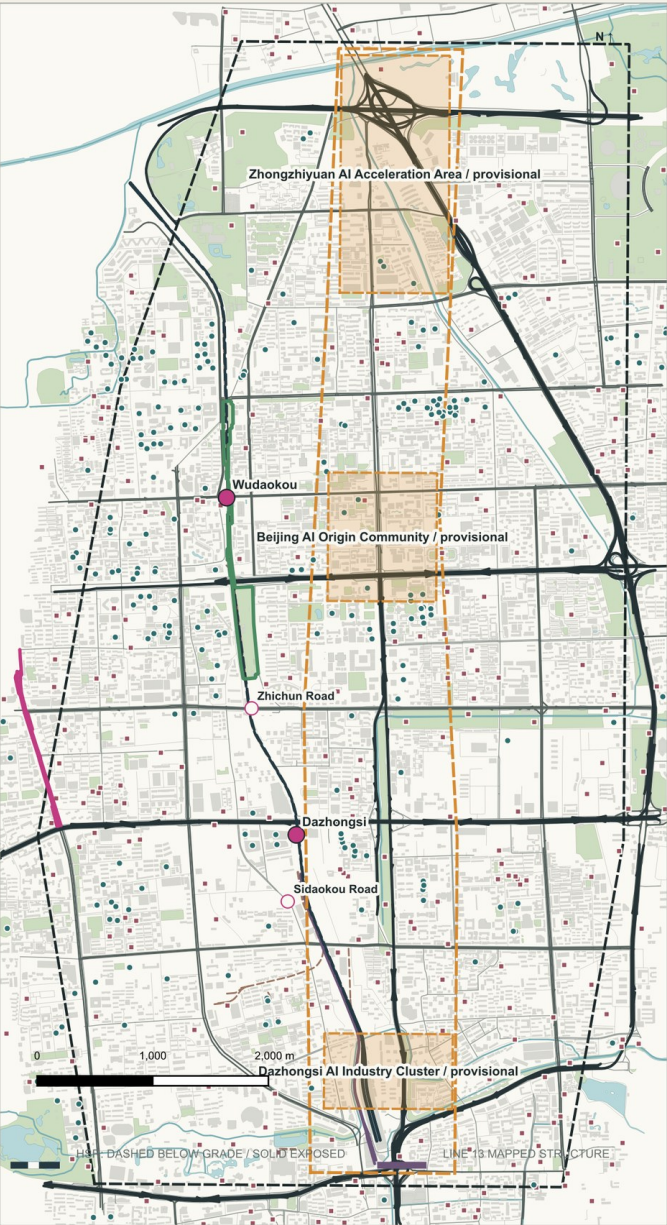
Public fact, mapped orientation, provisional extent, and unknowns remain separate

01 / 43.6 km² STRATEGIC DIAGNOSIS

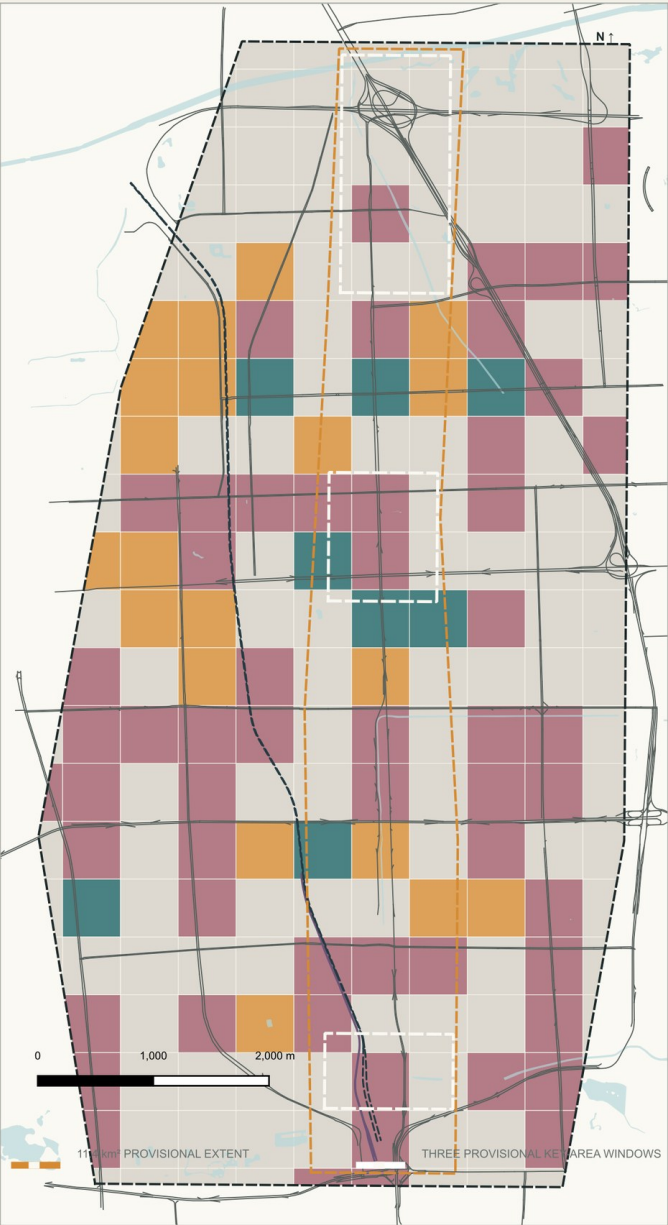
CONCENTRATED INNOVATION, UNEVEN EVERYDAY SUPPORT

2024 PUBLIC-DRAFT CONTEXT * 2026 PROVISIONAL EXTENT * OSM 500 m ORIENTATION GRID / CONCEPT—NOT AN OFFICIAL REDLINE

A REAL URBAN FABRIC AND THREE EXTENT TIERS



B 500 m PUBLIC-MAP SIGNALS (NOT A NEEDS ASSESSMENT)



C EVIDENCE-SUPPORTED JUDGEMENT



LIMIT: extent polygons are not official redlines; OSM completeness varies. Grid signals do not establish need, capacity, ownership, permission, or engineering feasibility.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Spatial structure: one spine, Three Zones, Two Wings, three testbeds

Repair the ordinary city first; trial only inside bounded, recoverable units

02 / 11.4 km² OVERALL SPATIAL STRUCTURE

SPATIAL FRAMEWORK: AI FROM LAB TO CITY

REAL FABRIC × ONE SPINE × THREE ZONES × TWO WINGS × THREE TESTBEDS / FUNCTIONAL ANCHORS—NOT OFFICIAL KEY-AREA BOUNDARIES

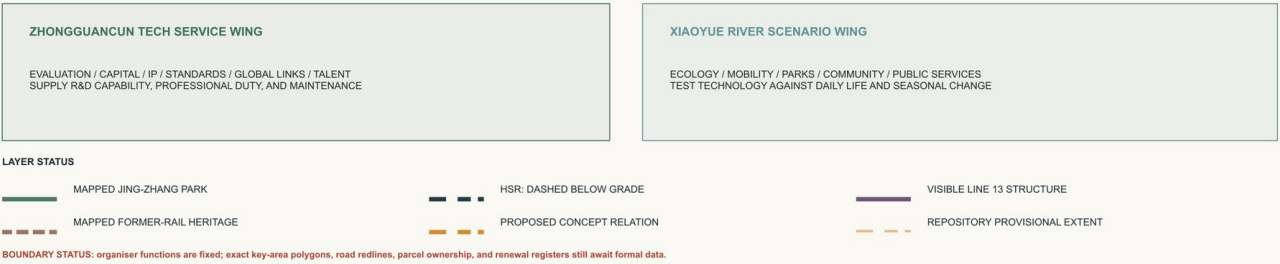
A CONCEPT STRUCTURE ON MAPPED FABRIC



B THREE ZONES × THREE TESTBEDS: FUNCTIONAL ANCHORS, NOT INVENTED REDLINES

01	ZHONGZHIYUAN AI ACCELERATION AREA	INTERFACES
	TECHNOLOGY TESTBED	JZ-07 CAU-R&D INTERFACE · JZ-08 XIAOYUE-QINGHE
	FUNCTION	SPATIAL ACTION
	FULL-STACK R&D → SAFETY GOVERNANCE → CONTROLLED VALIDATION	PARK-WATER-INDUSTRY INTERFACE; HUMAN TRIALS MUST BE ISOLATABLE, TAKE-OVER-READY, AND RECOVERABLE.
02	BEIJING AI ORIGIN COMMUNITY	INTERFACES
	INNOVATION TESTBED	JZ-03 XITUCHENG · 04 QINGHUAYUAN · 05 LIJDAOKOU · 06 QINGHUA EAST RD
	FUNCTION	SPATIAL ACTION
	LAB → OPEN SOURCE → INCUBATION → REAL-PROBLEM FEEDBACK	REPAIR CAMPUS-PARK-STREET THRESHOLDS, WALKING CONTINUITY, AND LOW-DISRUPTION ADAPTIVE REUSE.
03	DAZHONGSI AI INDUSTRY CLUSTER	INTERFACES
	URBAN LIFE TESTBED	JZ-01 BEIJING NORTH-COMMUNITY · JZ-02 DAZHONGSI-PARK-MUSEUM
	FUNCTION	SPATIAL ACTION
	PRODUCT → BUSINESS → PUBLIC LIFE → HUMAN FEEDBACK	CONNECT STATION-COMMERCE-PARK-COMMUNITY; FOUR-QUADRANT ENGINEERING AWAITS FORMAL DATA.

TWO WINGS SUPPLY CAPABILITY AND CONTEXT—NOT DEVELOPMENT BOUNDARIES



C VALUE FLOW

HUMAN FEEDBACK RETURNS TO THE RESEARCH QUESTION; ANY FAILED HARD GATE STOPS OR EXITS THE TRIAL.



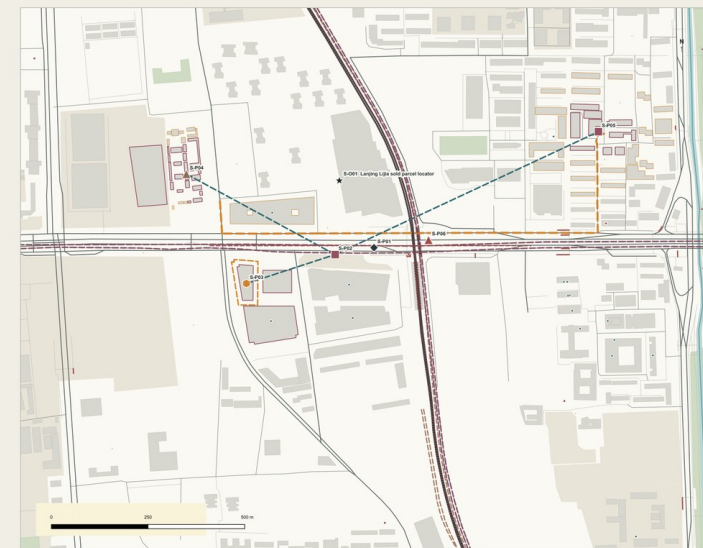
Sources: 2026 official competition announcement and public updates; repository provisional geometry; OpenStreetMap contributors (ODbL); official rail/park context. CRS EPSG:4548. Concept proposal for review — not an approval or implementation plan.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL



S DAZHONGSI | A NO-PHONE DAILY CHAIN FROM FOUR-QUADRANT TRANSFER TO HUMAN SERVICE

72.8 ha boundary pending / Line 12 • HDR tunnel / Line 13 embankment • elevated station



ON PROJECT SEARCHES (ALL SUBJECT TO APPROVAL)		
	INITIAL ACTION	DELIVERY DATE
0.001	Dishwasher / Four-Quadrant Washing Boat Residue, chemicals, vertical access, washing and cycle parking	Finalised layout, colour coordination, entrance, transport route
0.002	Boston Human Help Welfare No phone guidance and assisted (boat with human involvement)	Washing, container storage, assembly, privacy, emergency
0.003	Enterprise Controlled Ground Floor Lab New product tests inside a portable ground floor unit with 3 types	Owner, the building, site, noise and mechanism
0.004	Swimming Bathing Boat + Water Park New water, hot tubs and water slides, water slides, water slides	Non-toxicity, hygiene, safety, outdoor access
0.005	Community Health Human Referral Guidance, accommodation, services and human referral only	Initiation, build, delivery, privacy, washing, emergency
0.006	Auto Wash + Cycle Management Auto (high parking, low level, delivery, parking and walking control)	Flow sheet, operation, vehicle, initiation, washing, pool

 MAPPED RETRO-RAIL, HERITAGE

FIG 85 V6 / 2024-09-18 / CONCEPT—NOT AN APPROVAL.

05/14

Zhongzhiyuan: Technology Testbed

Keep controlled trials inside; repair ordinary movement gaps first

03-N

ZHONGZHIYUAN | KEEP CONTROLLED TRIALS INSIDE; REPAIR ORDINARY MOVEMENT FIRST

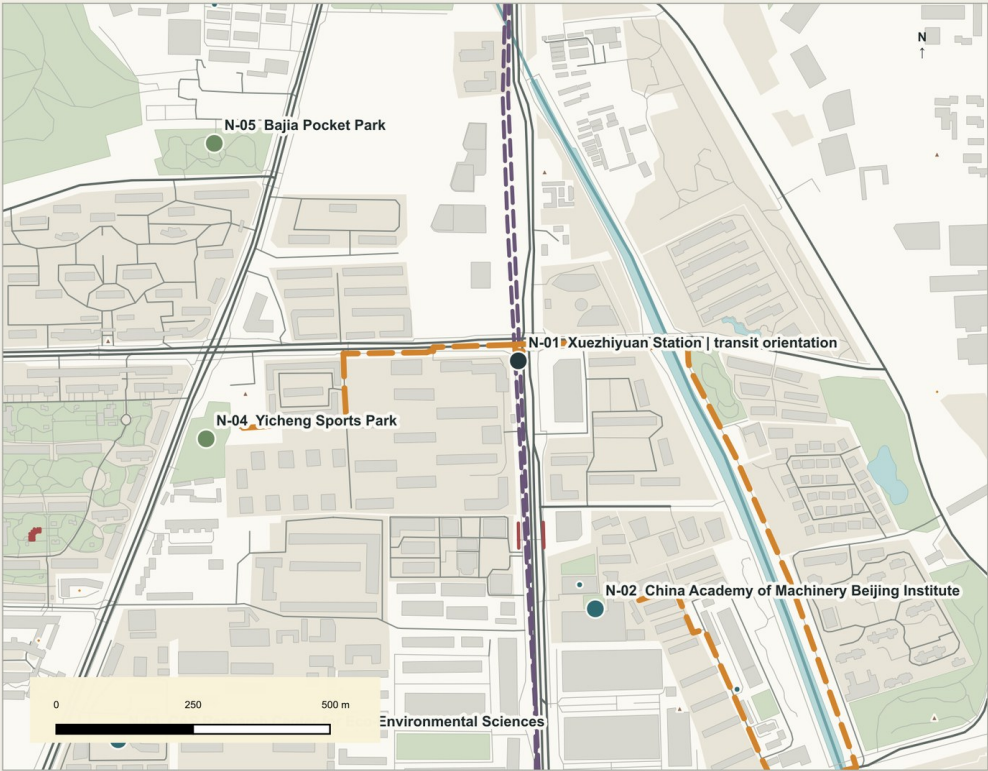
192.1 ha detailed boundary pending / mapped window 1.8 × 1.4 km / both maps 1:10000

A3 REVIEW V1

PROVISIONAL / NO OFFICIAL BOUNDARY

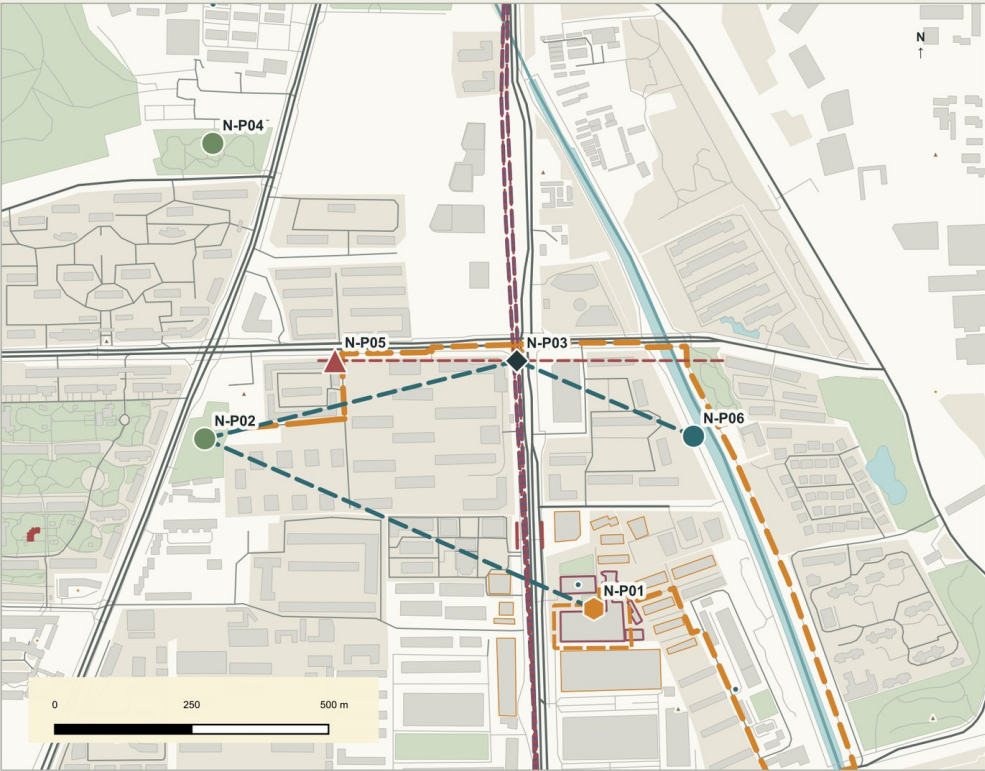
EXISTING / PUBLIC-MAP ORIENTATION

mapped ≠ verified



PROPOSED / CONCEPT OVERLAY

6 concepts - survey A5/B15



PROJECT REGISTER / SIX SEARCHES AND KEY GATES

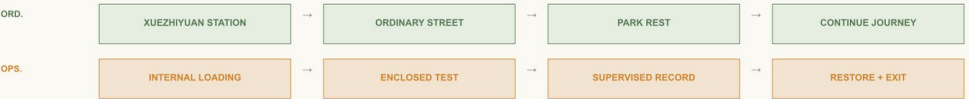
	PROJECT	KEY GATE
N-P01	Controlled Test Court	Owner · fire · loading
N-P02	Yicheng Park Recovery Point	Opening · accessible paths · operator
N-P03	Xuezhiyuan Ordinary Arrival	Entrance · footfall · vertical access
N-P04	Bajia Pocket Park Quiet Rest	Site edge · trees · lighting
N-P05	East–West Break Audit Gate	Field walk · ownership · road authority
N-P06	Xiaoyue Waterfront Interface Check	River authority · flood · ecology

SECTION / OPERATING LOGIC · ELEVATION UNSURVEYED



TWO LANES / ORDINARY USERS SEPARATE FROM TEST OPERATIONS

NO FORCED PARTICIPATION



OSM orientation / EPSG:4548 / official locator ≠ parcel / test envelope ≠ mapped building + 12m graphic buffer / demolish-retain UNASSIGNED

CONCEPT—NOT AN APPROVAL

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Beijing AI Origin: Innovation Testbed

Build an Origin–heritage–street learning chain independent of campus access

03-M

AI ORIGIN | AN ORIGIN–HERITAGE–STREET LEARNING CHAIN INDEPENDENT OF CAMPUS ACCESS

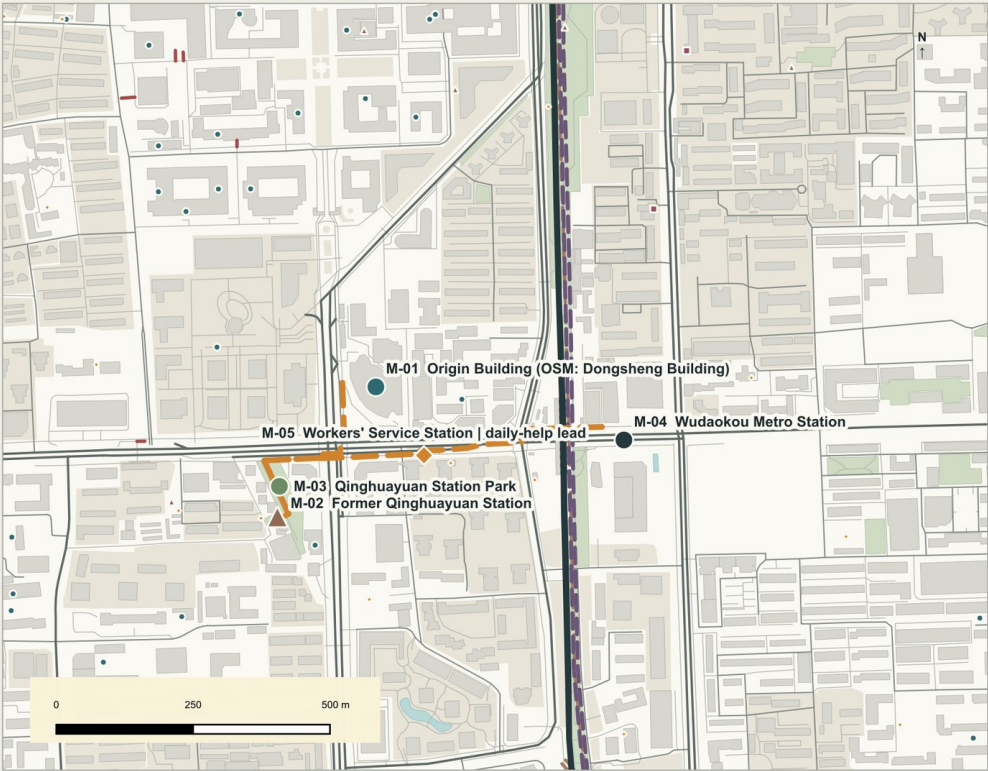
A3 REVIEW V1

PROVISIONAL / NO OFFICIAL BOUNDARY

104.3 ha detailed boundary pending / mapped window 1.8 × 1.4 km / both maps 1:10000

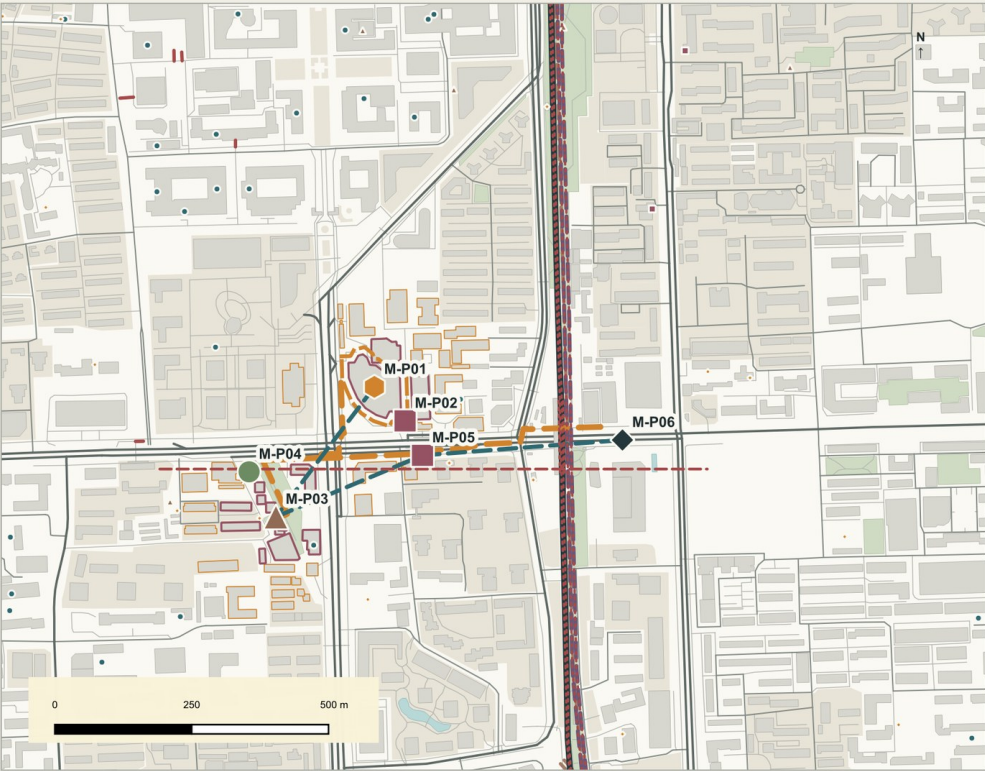
EXISTING / PUBLIC-MAP ORIENTATION

mapped ≠ verified



PROPOSED / CONCEPT OVERLAY

6 concepts - survey A15/B37



PROJECT REGISTER / SIX SEARCHES AND KEY GATES

	PROJECT	KEY GATE
M-P01	Origin Controlled Ground-Floor Lab	Building · fire · power
M-P02	Origin Human Help Desk	Ground-floor access · staffing · language
M-P03	Former Station No-Screen Learning Walk	Heritage edge · access · exhibit permit
M-P04	Station Park Group Learning Table	Trees · paths · lighting
M-P05	Workers' Station Daily Help Point	Existing service · access · staffing
M-P06	Wudaokou Campus–Park–Street Threshold	Station footfall · crossing · campus gates

SECTION / OPERATING LOGIC · ELEVATION UNSURVEYED

LINE 13 VIADUCT MAPPING

ORIGIN / FORMER STATION

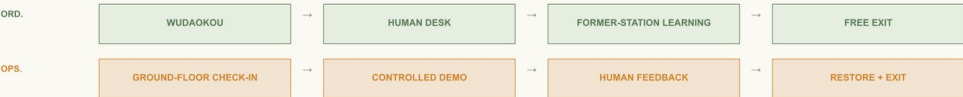
WUDAOKOU

JING-ZHANG HSR BELOW GRADE

SCHEMATIC / DEPTH / STRUCTURE / PROTECTION DISTANCE

TWO LANES / ORDINARY USERS SEPARATE FROM TEST OPERATIONS

NO FORCED PARTICIPATION



OSM orientation / EPSG:4548 / official locator ≠ parcel / test envelope ≠ mapped building + 12m graphic buffer / demolish-retain UNASSIGNED

CONCEPT—NOT AN APPROVAL

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Dazhongsi: Urban Life Testbed

A no-phone station-to-service journey; non-diagnostic and human-accountable

03-S

DAZHONGSI | A NO-PHONE DAILY CHAIN FROM FOUR-QUADRANT TRANSFER TO HUMAN SERVICE

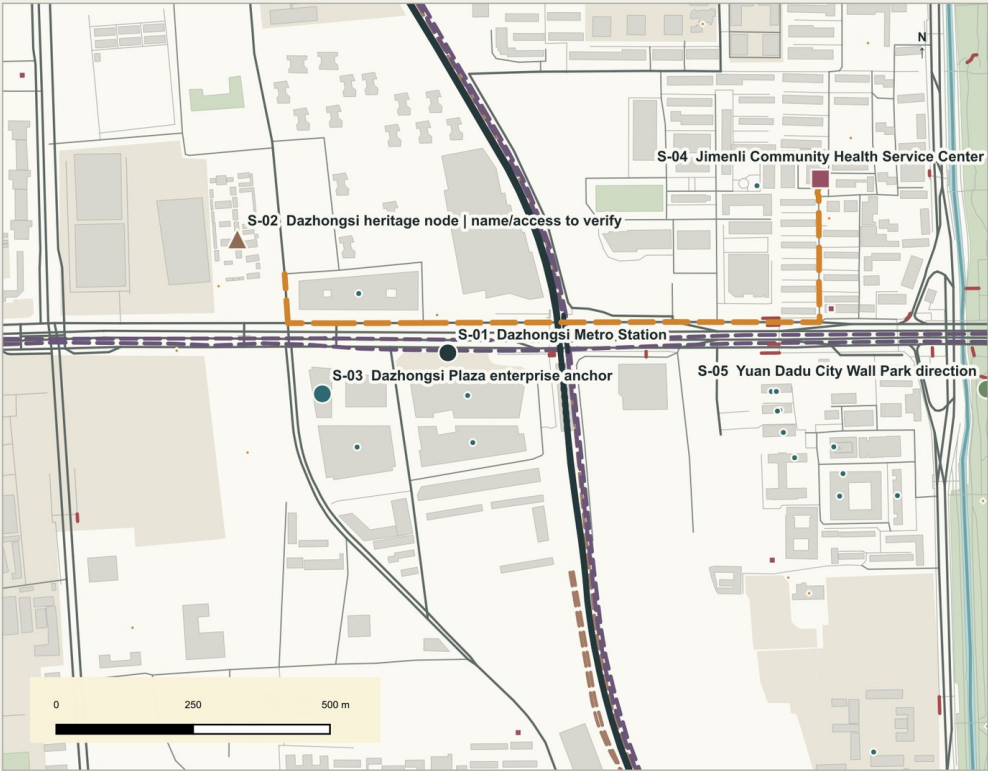
A3 REVIEW V1

PROVISIONAL / NO OFFICIAL BOUNDARY

72.0 ha detailed boundary pending / mapped window 1.8 × 1.4 km / both maps 1:10000

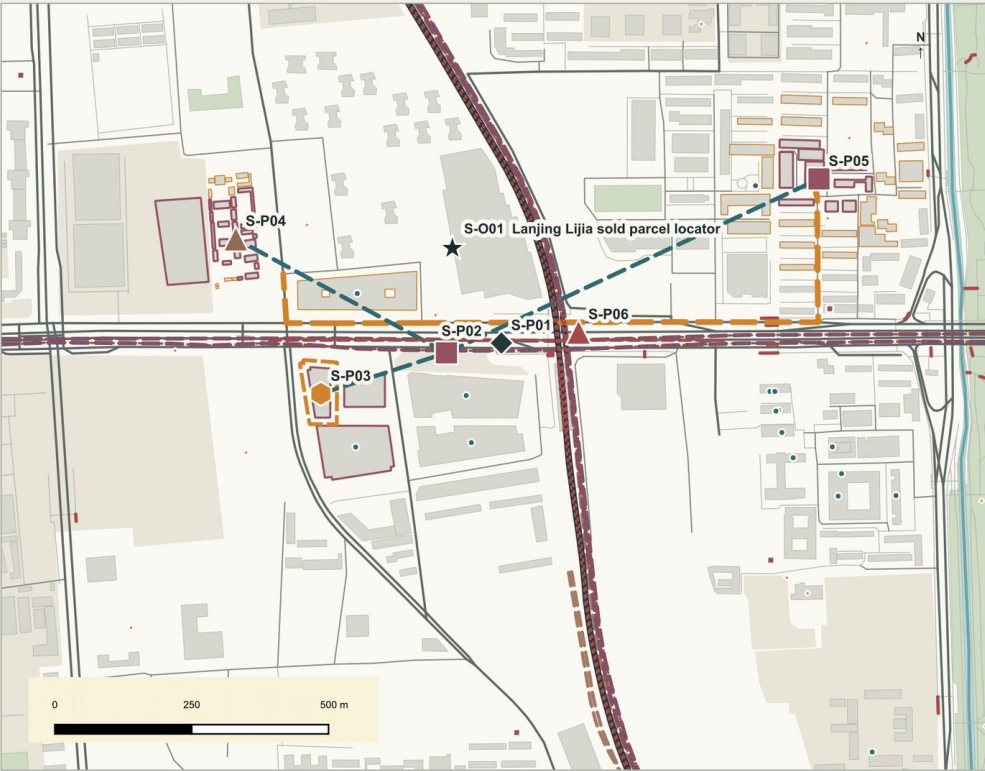
EXISTING / PUBLIC-MAP ORIENTATION

mapped ≠ verified



PROPOSED / CONCEPT OVERLAY

6 concepts + 1 official locator · A33/B32



PROJECT REGISTER / SIX SEARCHES AND KEY GATES

	PROJECT	KEY GATE
S-P01	Dazhongsi Four-Quadrant Walking Knot	Footfall · signals · columns/utilities
S-P02	Station Human Help Window	Staffing · operator approval · accessibility
S-P03	Enterprise Controlled Ground-Floor Lab	Owner · fire · loading
S-P04	Dazhongsi Heritage Rest + Water Point	Name/access · heritage · capacity
S-P05	Community Health Human Referral	Institution · clinical boundary · privacy
S-P06	Station Kerb + Cycle Management	Time-sliced observation · redline · enforcement

SECTION / OPERATING LOGIC · ELEVATION UNSURVEYED

LINE 13 EMBANKMENT / ELEVATED STATION

GROUND FLOOR

HUMAN HELP / COMMUNITY

LINE 12 BELOW GRADE

SCHEMATIC # DEPTH / STRUCTURE / PROTECTION DISTANCE

TWO LANES / ORDINARY USERS SEPARATE FROM TEST OPERATIONS

NO FORCED PARTICIPATION

ORD.

METRO ARRIVAL

HUMAN HELP

HERITAGE REST

COMMUNITY REFERRAL

OPS.

GROUND FLOOR

DEVICE TEST

EXCEPTION → HUMAN

RESTORE + EXIT

OSM orientation / EPSG:4548 / official locator ≠ parcel / test envelope = mapped building + 12m graphic buffer / demolish-retain UNASSIGNED

CONCEPT—NOT AN APPROVAL

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Mobility, rail verticality, and blue-green public space

Where no lawful crossing exists, the concept line stops at the operating boundary

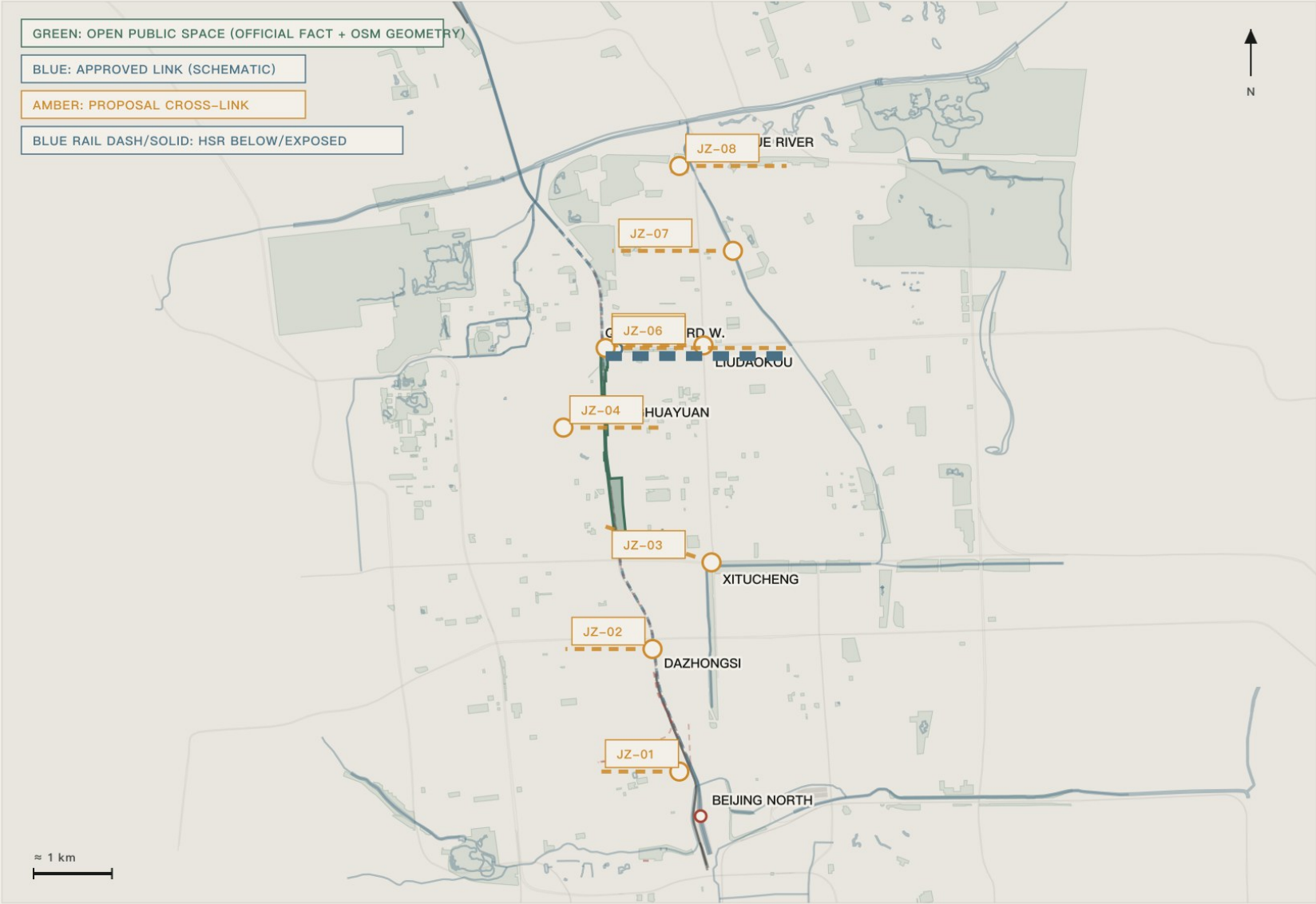
JZ-RCT / 04 / V0.1

04 / REAL PUBLIC SPINE × CROSS-BLUE-GREEN NETWORK

STATUS-AWARE NETWORK

NOT ONE GLOWING AI LINE A STATUS-AWARE PUBLIC NETWORK

Distinguish open, approved and merely mapped segments.
Then use eight cross-links to organise
station–park–campus–street–service journeys.



FIVE COMPLETE JOURNEYS

- 01 **STATION → PARK → SERVICE**
STATIC WAYFINDING / REST / CROSSING / HUMAN HELP
- 02 **CAMPUS → PUBLIC EDGE → COMMUNITY**
RESEARCH ACCESS IS NOT HIDDEN BEHIND CAMPUS ID
- 03 **ORDINARY PATH // BOUNDED TRIAL**
REFUSAL DOES NOT COST ACCESS OR SERVICE
- 04 **JING-ZHANG → COLLEGE RD → XIAOYUE**
APPROVED LINK AND PROPOSAL TRIAL STAY DISTINCT
- 05 **SERVICE → RETURN**
NO CONTINUOUS TRACKING; ONE-TIME, CANCELLABLE HELP

DRAWING DISCIPLINE

NO CONTINUOUS OPEN EVIDENCE → NO CONTINUOUS GREEN SPINE

NO FIELD CHECK → NO ACCESSIBILITY COVERAGE RATE

NO PERMIT → NO ROBOT IN PUBLIC RIGHT-OF-WAY

Basemap and park geometry: OpenStreetMap contributors, 2026–08–10. Open/approved status: official Beijing and Haidian sources. Cross-links are concepts, not road or engineering alignments.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Flagship 01: AI Learning Corridor

Problem Desk, no-screen explanation, Reverse Lab, Failure and Revision Library

JZ-RCT / SCENE-LEARNING / V0.1

FLAGSHIP 01 / AI LEARNING CORRIDOR

THE CITY ASKS FIRST. THE LAB ANSWERS SECOND.

A human hosts the problem desk; AI only offers optional grouping and explanation.
People can submit, understand and reach a human without a login or QR code.

AI-GENERATED CONCEPT SCENE / NOT A SITE PHOTO / NOT AN APPROVAL



SPATIAL RELATION

- 1 CONTINUOUS ORDINARY PATH + SHADE
- 2 HUMAN-HOSTED PROBLEM DESK
- 3 OPTIONAL, SWITCHABLE AI TERMINAL
- 4 RAIL HERITAGE + PUBLIC LEARNING SPACE

REAL TASK

Let people confirm the real urban problem instead of allowing a technical team to rewrite it for them.

STOP NOW

Stop if a request is distorted, identity leaks, urgency is downgraded, or no human owns the response.

WHAT REMAINS WITHOUT AI

Paper problem desk, places to sit, no-screen explanation and a public failure/revision archive.

NOW: V0.1 / NO FIELD TRIAL

Concept image explains spatial and role relationships only. It is not existing condition, site permission, partnership, device selection or final design.
Generated with built-in OpenAI image generation. Exact copy, data and boundaries remain deterministic vector overlays.

LOCAL CANDIDATE / DO NOT SUBMIT
BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Flagship 02: Embodied AI Urban Lab

Enclosed and slow; people first; physical stop; human takeover; restoration

JZ-RCT / SCENE-EMBODIED / V0.1

FLAGSHIP 02 / EMBODIED AI URBAN LAB

LEARN TO YIELD. LEARN TO FAIL SAFELY.

An ordinary path runs beside a bounded test cell.
On-site people take over every failure; passers-by are not test conditions.

AI-GENERATED CONCEPT SCENE / NOT A SITE PHOTO / NOT AN APPROVAL



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SPATIAL RELATION

- 1 ORDINARY PATH NEVER OCCUPIED
- 2 LOW-SPEED BOUNDED TEST CELL
- 3 ON-SITE MARSHAL + PHYSICAL STOP
- 4 RECOVERY BAY OFF THE PUBLIC PATH

REAL TASK

Test whether a device yields, degrades, stops and can be taken over — not merely whether it can move.

STOP NOW

Stop on any injury/near miss, ordinary-path incursion, failed emergency stop or retained identity data.

WHAT REMAINS WITHOUT AI

Clearer right-of-way, handoff and recovery rules, plus public near-miss and takeover records.

NOW: V0.1 / NO FIELD TRIAL

LOCAL CANDIDATE / DO NOT SUBMIT
BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Flagship 03: AI Health and Aging Lab

No-phone service finding, rest-to-service routes, caregiver human handoff

JZ-RCT / SCENE-HEALTH / V0.1
FLAGSHIP 03 / AI HEALTH & AGING LAB

COMPLETE THE JOURNEY WITHOUT A SMARTPHONE.

Seat, shade, water, static cues and a staffed window come first.
AI explains verified services only: no diagnosis, score or continuous tracking.

AI-GENERATED CONCEPT SCENE / NOT A SITE PHOTO / NOT AN APPROVAL



Concept image explains spatial and role relationships only. It is not existing condition, site permission, partnership, device selection or final design.
Generated with built-in OpenAI image generation. Exact copy, data and boundaries remain deterministic vector overlays.

SPATIAL RELATION

- 1 SEAT, WATER + STEP-FREE PATH
- 2 STATIC DIRECTION, NO PHONE
- 3 ACCOUNTABLE HUMAN SERVICE WINDOW
- 4 OPTIONAL ONE-SESSION VOICE KIOSK

REAL TASK

Complete finding, reaching, resting, human handoff and return — not simply add a smart interaction.

STOP NOW

Stop on high-risk misdirection, unnecessary identity/medical data, unavailable human help or continuous location.

WHAT REMAINS WITHOUT AI

Verified paper directory, a rested route, static wayfinding, human help and a clear return path.

NOW: V0.1 / NO FIELD TRIAL

LOCAL CANDIDATE / DO NOT SUBMIT
BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Twelve scenarios × City Release

Every card begins with ordinary baseline, permission, stop, exit, and public return

SC01

City Problem Desk

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC02

No-Screen Explanation Walk

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC03

Reverse Lab

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC04

Failure and Revision Library

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC05

Shared Right-of-Way Sandbox

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC06

Curb Handoff Bay

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC07

Failure and Takeover Drill

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC08

Non-Identifying Public-Realm Check

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC09

No-Phone Service Finder

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC10

Rest-to-Service Route

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC11

Caregiver Human Handoff

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC12

Service Directory Audit

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

Metrics, evidence, unknowns, and next actions

Recomputable does not mean official; FAR, height, density, and field performance remain unknown

JZ-RCT / 05 / V0.1

05 / INNOVATION × DAILY LIFE: EVIDENCE DECIDES WHAT COMES FIRST

RELATIVE PROXY / NOT OFFICIAL

157 HIGH-I × LOW-LIFE CELLS
MEAN FIELD CHECKS, NOT AI PROJECTS

These are field-verification candidates
under current public mapped proxies.
Ordinary-city repair first; conditional AI second.



EVIDENCE LEDGER

906

STUDY-WINDOW CELLS

≈200–250 m samples; not parcels

587 / 38 / 281

HIGH / MED / LOW CONFIDENCE

Low confidence is greyed in formal figures

98

INNOVATION NETWORK NODES

176 internal/semantic items excluded

40

OFFICIAL NEAR-CORRIDOR ELDER STATIONS

Unverified coordinates; excluded from score

DIAGNOSIS → SPATIAL ACTION

HIGH-I × LOW-LIFE: VERIFY ORDINARY SERVICES; DO NOT AUTO-DEPLOY AI;
LOW-I × LOW-LIFE: FILL EVIDENCE GAPS; UNKNOWN IS NOT ZERO;
K1 conflicts with its label; K2 varies internally; K3 geometry drifts;
EVERY TESTBED NEEDS FIELD, PERMISSION, OPERATOR AND EXIT RESOURCE.

MAIN: S0 / 800 M / DL4

SENSITIVITY: S1 / 400–1200 M / DL5

Data: OpenStreetMap contributors, common baseline 2026-08-10T03:51:58Z; official Haidian elder-service lists. Median-relative quadrants are not official service, capacity, quality or need assessments.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL